

ENVIRONMENTAL IMPACT ASSESSMENT



EN302 — Environmental Impact Assessment & Audit

GROUP - 6 ECOLOGY AND GREEN COVER

Project	Construction of Academic Block-1 (AB-1), DTU Delhi
Location	Delhi Technological University, Shahabad Daulatpur, Bawana Road, North-West Delhi
EIA Category	Category B1 — Educational Institutions >20,000 sqm built-up area
Survey Area	11,397.07 sqm (Area 1) 12,217.89 sqm (Area 2)
Total Trees	198 trees 15 species 17 zones
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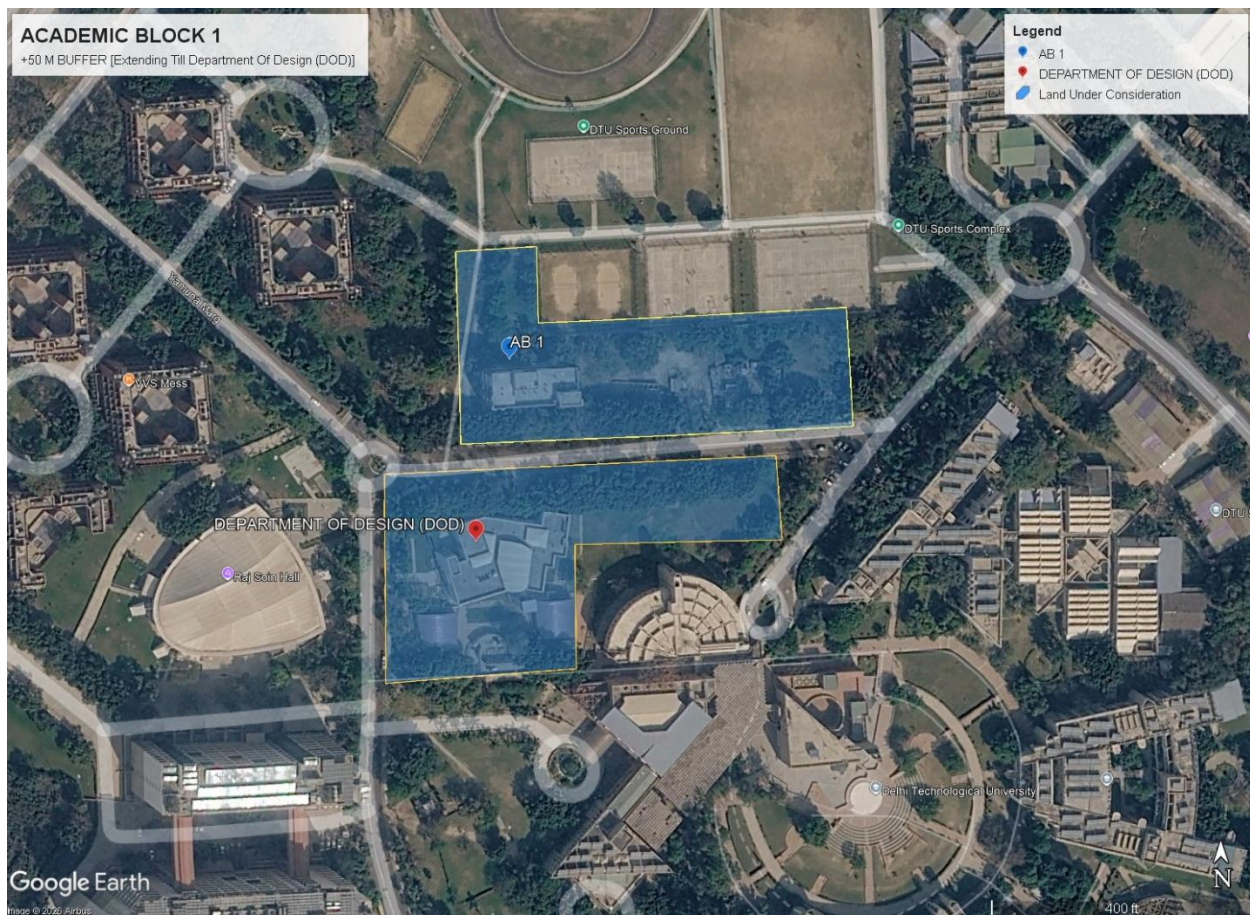
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Abstract

This report presents the findings of the Ecology and Green Cover baseline study conducted as part of the Environmental Impact Assessment (EIA) for the proposed construction of Academic Block-1 (AB-1) at Delhi Technological University (DTU), Shahabad Daulatpur, North-West Delhi. The study is mandated under EIA Notification 2006 (Category B1) and covers five key tasks assigned to Group 6: tree census and inventory, green cover loss quantification, faunal survey, biodiversity index analysis, and green belt and mitigation planning.

A total of 198 trees belonging to 15 species were enumerated across 17 survey zones spanning two areas of the campus. The dominant species is Banyan (*Ficus benghalensis*), consistent with DTU's heritage urban forest character. The Shannon-Wiener Diversity Index (H') was calculated at 2.48, indicating moderate-to-good biodiversity for an urban institutional campus. Faunal surveys recorded 49 species across both areas including birds, mammals, reptiles, amphibians, and invertebrates.

Green cover loss analysis using Google Earth measurements reveals that the proposed AB-1 construction (1,006.05 sqm building footprint + associated cemented and fallow areas totalling 3,424.05 sqm) has resulted in an estimated loss of 80 trees from a pre-construction baseline of approximately 182 trees. As per DTU's compensatory afforestation policy and MoEFCC guidelines, 800 trees must be planted (1:10 ratio) across campus, requiring approximately 2.83 hectares. A comprehensive mitigation plan including green belt design, transplantation, and monitoring schedule has been proposed.



1. Introduction

1.1 Project Background

Delhi Technological University (DTU), located in Shahabad Daulatpur, Bawana Road, North-West Delhi, is one of India's premier technical universities spread across a 164-acre green campus. The university has proposed the construction of Academic Block-1 (AB-1) as part of its Master Plan to expand academic infrastructure. The project involves a ground coverage of 1,006.05 sqm with G+9 floors, reaching a maximum building height of 36.45 m, resulting in a total built-up area of approximately 10,060.5 sqm.

The campus is situated within the Indo-Gangetic alluvial plains amidst rapid urbanisation, bordered by Rohini residential zones (Zone M) to the south and Bawana industrial areas (Zone P-Narela) to the north. The site's flat terrain, humid subtropical climate (summer highs of 39°C, winter lows of 7°C), and its proximity to Samaypur-Badli and Rithala Metro Stations raise critical environmental concerns that necessitate a formal EIA.

1.2 EIA Framework & Legal Basis

The proposed AB-1 construction falls under the purview of the EIA Notification 2006 (as amended) issued by the Ministry of Environment, Forest and Climate Change (MoEFCC), Government of India. Being an educational institution with a total built-up area exceeding 20,000 sqm, the project is classified as Category B1, requiring State Environment Impact Assessment Authority (SEIAA) clearance.

Applicable Legal Framework

EIA Notification 2006 (MoEFCC) | Delhi Preservation of Trees Act, 1994 | National Green Tribunal (NGT) Orders | Delhi Building Bye-Laws | NBC 2016 | MoEFCC Compensatory Afforestation Guidelines | Construction & Demolition Rules 2016

1.3 Scope of This Chapter

This chapter specifically addresses the ecological baseline of the proposed AB-1 site and its 50m buffer zone within the DTU campus. The study encompasses:

- Task 1: Tree Census and Zone-wise Inventory (198 trees, 15 species, 17 zones)
- Task 2: Green Cover Loss Quantification with applicable laws
- Task 3: Faunal Survey using transect method (49 species across 2 areas)
- Task 4: Biodiversity Index Analysis (Shannon-Wiener, Simpson's, Evenness, Margalef's)
- Task 5: Green Belt Design and Mitigation Plan with compensatory afforestation

2. Methodology

2.1 Study Area Delineation

The study area was delineated using Google Earth Pro satellite imagery. The campus was divided into two major areas — Area 1 (AB-1 site, 11,397.07 sqm) and Area 2 (DOD, 12,217.89 sqm) — each further subdivided into smaller survey zones using a grid-based approach. Area 1 comprises 9 sub-zones (A1 to I1) and Area 2 comprises 8 sub-zones (A2 to H2), resulting in 17 total survey zones.

2.2 Tree Inventory Method

Trees were enumerated using the complete enumeration method within each zone. For each tree, the following parameters were recorded: species (common and botanical name), Girth at Breast Height (GBH) measured at 1.37 m above ground level using a measuring tape (in inches), health condition, and approximate location within the zone. Species not directly observed but likely present based on campus ecology.

2.3 Faunal Survey — Transect Method

Faunal surveys were conducted using the line transect method across both survey areas. Each zone was treated as a transect unit. Species were identified through direct visual observation, acoustic identification (bird calls), and signs of presence (tracks, droppings, nests). Observations were recorded zone-wise along with locality information, species type (class), and ecological role.

2.4 Biodiversity Index Calculation

The following diversity indices were calculated using species abundance data from the tree census:

- *Shannon – Wiener Index* (H') = $-\sum(p_i \times \ln p_i)$, where $p_i = n_i/N$
- *Pielou's Evenness Index* (E) = H' / H_{max} , where $H_{max} = \ln(S)$
- *Simpson's Diversity Index* ($1 - D$) = $1 - \sum[n_i(n_i - 1) / N(N - 1)]$
- *Margalef's Species Richness Index* (DMg) = $(S - 1) / \ln(N)$

2.5 Green Cover Loss Quantification

Land cover within Area 1 was mapped and measured using Google Earth Pro. The area was classified into three categories: Green Cover (vegetative), Fallow Land (disturbed/bare), and Cemented Land (permanent structures). Pre-construction baseline was back-calculated by applying the current vegetative tree density to the full hypothetical pre-construction area. Loss was computed as the difference between the original estimated tree count and currently surviving trees.

Chapter 1: Tree Census & Zone-wise Inventory (Task 1)

1.1 Theoretical Background

A tree census is a systematic enumeration of all trees within a defined area, forming the foundational baseline for any ecological impact assessment. In the context of EIA, tree inventory data is essential for quantifying green cover, assessing biodiversity, predicting ecological impacts, and planning compensatory measures. The Census is guided by the principles of complete enumeration — every individual tree within the study boundary is counted and characterised.

Girth at Breast Height (GBH), measured at 1.37 m above ground, is the standard dendrometric parameter used in Indian forest surveys (as per Forest Survey of India protocols). GBH correlates with tree age, biomass, and carbon sequestration potential. The dominant presence of Banyan (*Ficus benghalensis*) on DTU campus is ecologically significant — Banyan trees are keystone species supporting birds, insects, and mammals through their large canopy and fruit production.

1.2 Part 1 — Zone-wise Tree Count

The survey area was divided into 17 zones across two areas. Area 1 (11,397.07 sqm) encompasses the AB-1 site region with 9 sub-zones, and Area 2 (12,217.89 sqm) covers the DOD grid with 8 sub-zones.

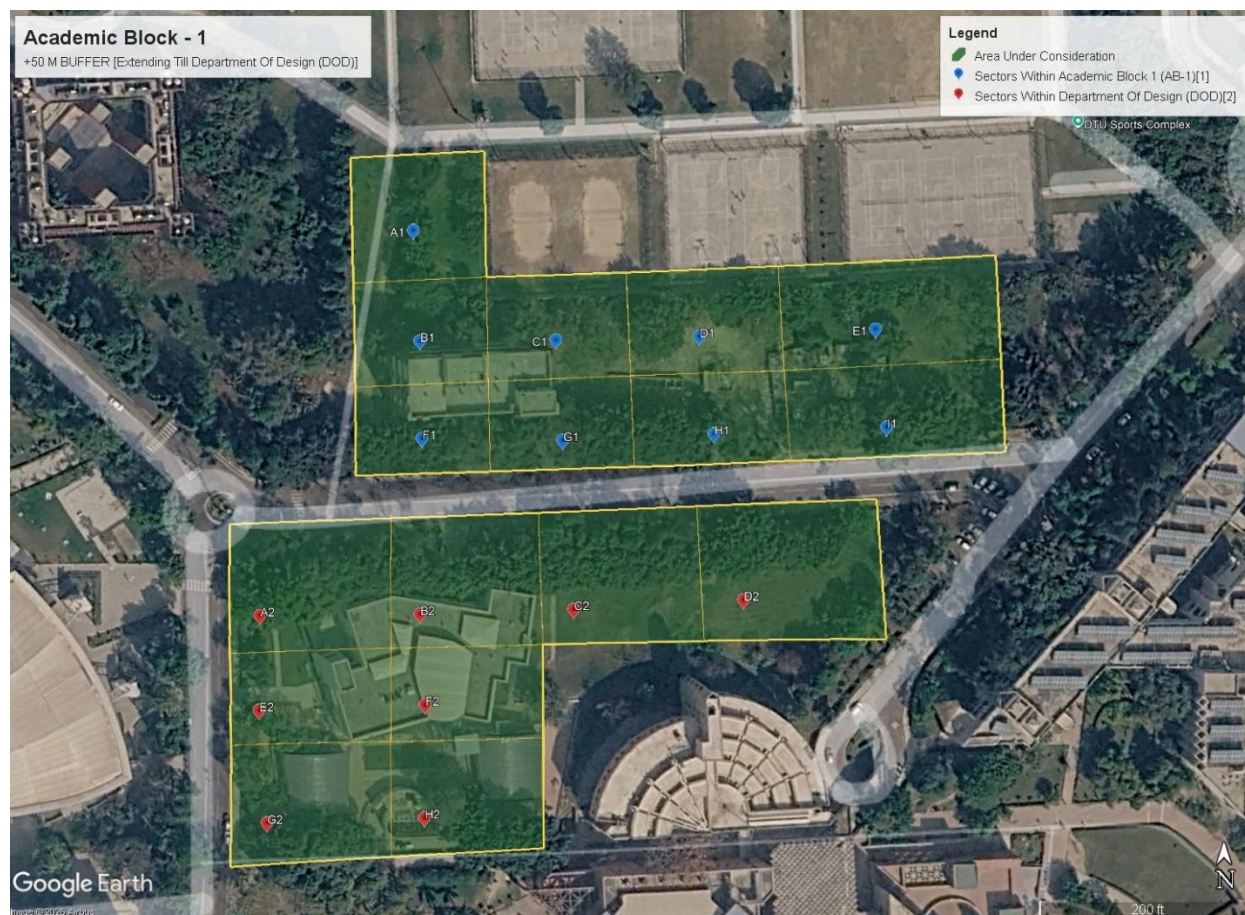


Table 1.1 —Zone-wise Tree Count For AB-1 And DOD (A1 to I1; A2 TO H2)

Zone	Area Description	Land Cover Type	Area (m ²)	No. of Trees Observed	% of Area Total
Academic Block-1 (AB-1)					
A1	North-west corner plot	Dense tree cover	1,328.91	14	13.7%
B1	West side – admin block adjacent	Mixed tree & built-up	1,115.34	8	7.8%
C1	Central-north strip	Moderate tree cover	1,191.06	12	11.8%
D1	Central strip	Dense tree cover	1,289.19	14	13.7%
E1	North-east strip	Dense tree cover	1,783.45	13	12.7%
F1	South-west corner	Mixed tree & built-up	967.87	8	7.8%
G1	South-central strip	Moderate tree cover	1,029.19	12	11.8%
H1	South-central strip	Moderate tree cover	1,131.01	11	10.8%
I1	South-east strip	Dense tree cover	1,561.05	10	9.8%
Sub-Total	Area 1 Total		11,397.07	102	
Department Of Design (DOD)					
A2	North-west green patch	Dense tree cover	1,692.33	18	18.8%
B2	Admin/building block	Built-up / sparse trees	1,557	6	6.3%
C2	Northern green strip	Dense tree cover	1,660.27	16	16.7%
D2	Water body & road strip	Sparse – paved/water edge	1,959.34	7	7.3%
E2	West side green patch	Moderate tree cover	1,220.36	12	12.5%
F2	Central building area	Built-up / sparse trees	1,140.98	5	5.2%
G2	South-west green corner	Dense tree cover	1,573.69	17	17.7%
H2	South-central patch	Moderate tree cover	1,413.92	15	15.6%
Sub-Total	Area 2 Total		12,217.89	96	
NET TOTAL	All Zones Combined		23,614.96	198	17 zones total

The above table presents a comprehensive zone-wise distribution of tree cover across the AB-1 and DOD areas of the DTU campus, highlighting variations in vegetation density, land use, and spatial extent. Each zone is categorized based on its dominant land cover type—ranging from dense and moderate tree cover to mixed and built-up areas—providing insight into how infrastructure influences green distribution. AB-1 shows relatively balanced tree presence with a mix of dense and moderate cover, whereas the DOD region exhibits slightly larger green patches with higher percentages in certain zones such as A2 and G2. The variation in tree counts and area percentages reflects the heterogeneity of landscape planning across zones. Overall, the table serves as a critical baseline for understanding spatial vegetation patterns, enabling informed decisions for biodiversity conservation, urban planning, and future green interventions on campus.

1.3 Part 2 — Species-wise Tree Detail with GBH

A total of 15 species were identified across the campus. Banyan dominates with 47 individuals (23.7%), followed by Peepal (21) and Neem (20). Fruit/flower-bearing and ornamental species are fewer in number, consistent with the natural ecology of a managed urban campus in the Indo-Gangetic plains.

Table 1.2 — Species-wise Detail: 15 Species, 198 Trees, GBH in Inches

S.No	Common Name	Botanical Name	Total Trees	% Share	Avg GBH (in)
1	Banyan	<i>Ficus benghalensis</i>	47	23.7%	56.5
2	Peepal	<i>Ficus religiosa</i>	21	10.6%	22.5
3	Neem	<i>Azadirachta indica</i>	20	10.1%	20
4	Shisham	<i>Dalbergia sissoo</i>	16	8.1%	50
5	False Ashoka	<i>Polyalthia longifolia</i>	13	6.6%	37
6	Arjun	<i>Terminalia arjuna</i>	13	6.6%	55
7	Mulberry	<i>Morus alba</i>	11	5.6%	32
8	Bottle Brush	<i>Callistemon citrinus</i>	9	4.5%	38
9	Ber / Jujube	<i>Ziziphus mauritiana</i>	9	4.5%	71
10	Amaltas	<i>Cassia fistula</i>	8	4.0%	30
11	Gulmohar	<i>Delonix regia</i>	8	4.0%	28
12	Eucalyptus	<i>Eucalyptus globulus</i>	6	3.0%	40
13	Jamun	<i>Syzygium cumini</i>	6	3.0%	35
14	Sagwan/Teak	<i>Tectona grandis</i>	4	2.0%	45
15	Pilkhan	<i>Ficus virens</i>	7	3.5%	60
Total	TOTAL: 15 Species		198	100%	42.28 (wtd)

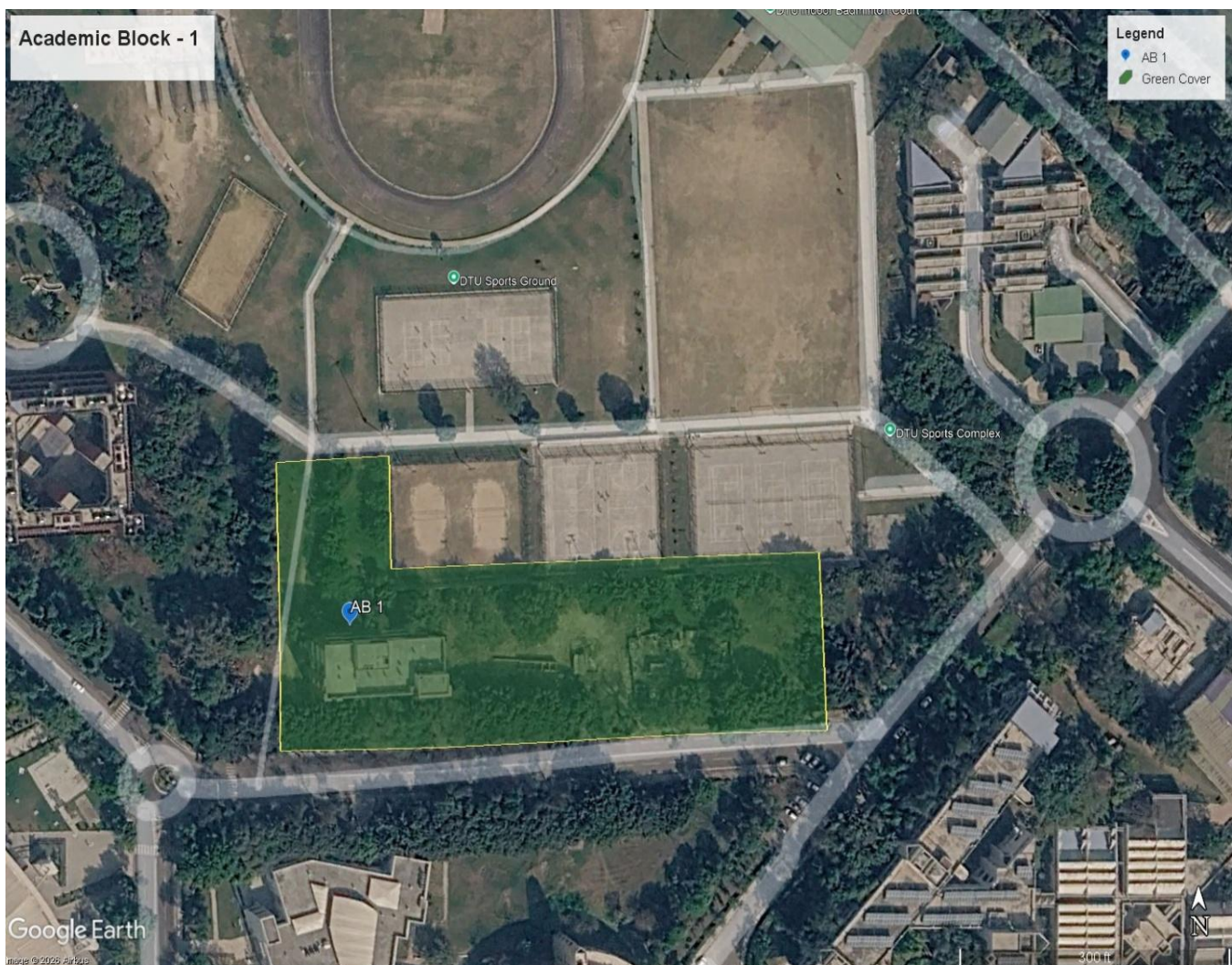
Note: GBH values for observed species are field-measured. The weighted average GBH across all species is 42.28 inches. Banyan trees show the widest GBH range (42–71 inches), reflecting both old heritage trees and newer plantations on campus.

Chapter 2: Green Cover Loss Quantification (Task 2)

2.1 Theoretical Background

Green cover loss quantification is a critical component of ecological baseline assessment in EIA. It involves mapping and measuring the pre-construction vegetative cover and comparing it against the post-construction state to determine the extent of ecological loss. In the Indian EIA framework, this is governed by the EIA Notification 2006, the Delhi Preservation of Trees Act 1994, and MoEFCC's Compensatory Afforestation Guidelines.

The methodology adopted here is a back-calculation approach: since construction has already occurred on the DTU campus (AB-1 and associated structures), the pre-construction baseline is reconstructed by applying the current vegetative tree density to the full hypothetical pre-construction area. This approach is consistent with EIA practice for projects where baseline conditions must be inferred from post-impact surveys.



2.2 Applicable Laws & Regulatory Framework

Table 2.1 — Applicable Legal Framework for Green Cover Loss

Act / Notification	Key Provision	Applicability	Implication for AB-1
Delhi Preservation of Trees Act, 1994	No tree shall be felled without prior written permission from the Tree Officer	Any tree felling within DTU campus	Compensatory plantation mandatory; permission required for each tree
EIA Notification 2006, MoEFCC	Baseline ecological data mandatory; green cover loss to be quantified and mitigated	Category B1 project — AB-1	Full tree inventory + mitigation plan required as part of EIA report
MoEFCC Compensatory Afforestation Guidelines	Minimum 1:10 tree replacement ratio for urban institutional projects	Trees lost due to AB-1 construction	800 trees to be planted; records maintained by DTU Estate Office
NGT Order (2019) — Urban Green Cover	No reduction in urban green cover without compensatory measures; heritage trees to be protected	DTU campus as urban green space	Banyan trees (GBH >50 in) must be protected; felling prohibited without NGT approval
Delhi Tree Act — Heritage Trees	Trees with GBH >100 cm (39 in) designated as heritage; protected from felling	Large Banyan trees on campus (GBH 56.5–71 in)	Heritage tree status likely applicable; mandatory conservation
C&D Waste Rules 2016, MoEFCC	Construction sites must have green screens/barriers to protect vegetation	AB-1 construction boundary	3m green barrier around construction zone mandatory
National Building Code 2016	Minimum 33% of site area to be maintained as green/open space	AB-1 footprint within campus	Post-construction landscaping mandatory within AB-1 precinct

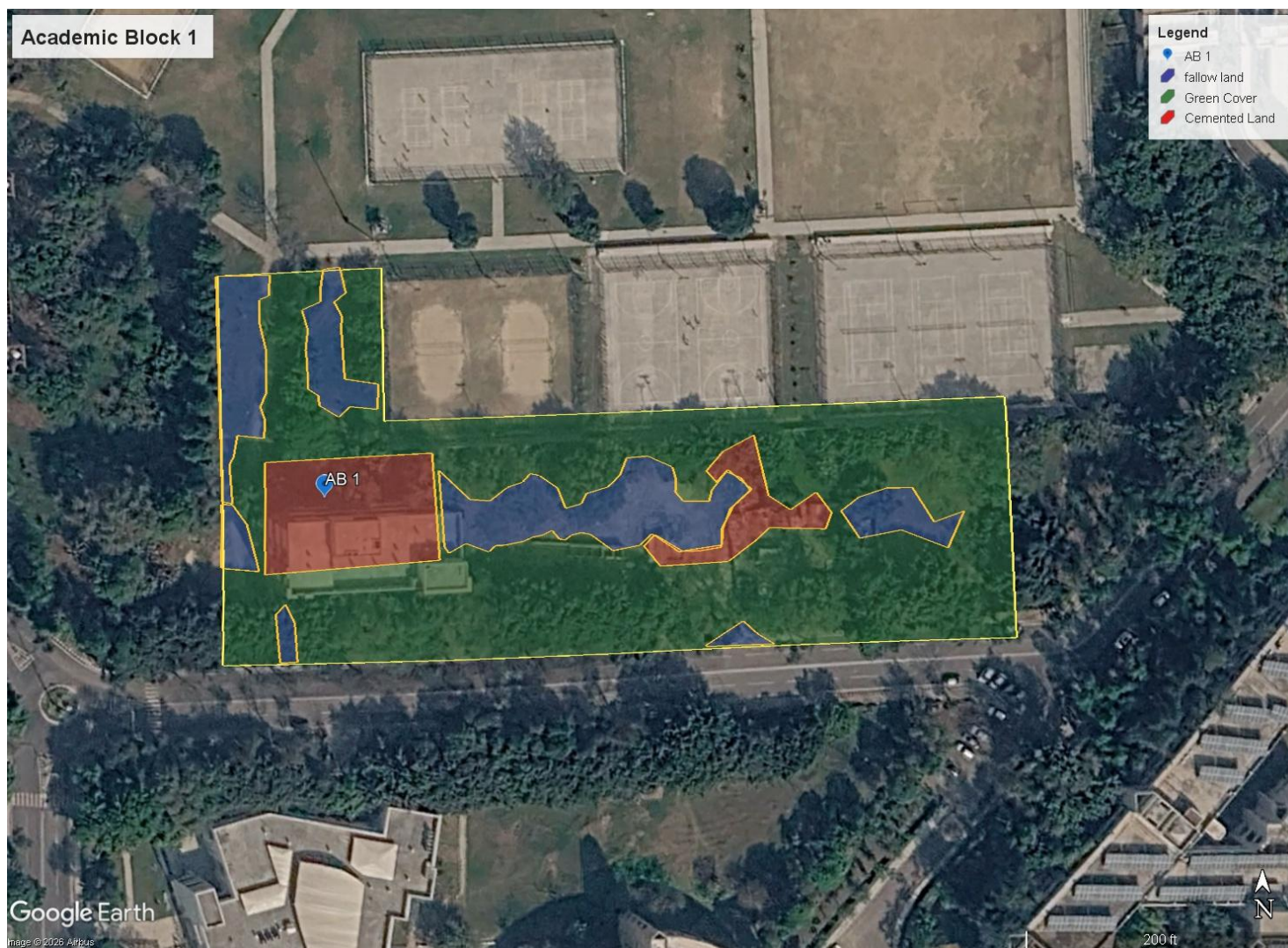
2.3 Land Cover Quantification — Area 1 [Academic Block-1] (Measured Data)

Land cover within Area 1 (11,397.07 sqm) was mapped and categorised using Google Earth Pro measurements. Three land cover categories were identified: Fallow Land (disturbed/bare ground), Cemented Land (permanent structures), and Green Cover (surviving vegetation).

Table 2.2 — Detailed Land Cover Breakdown (Google Earth Measurements)

Land Category	Sub-zone / Detail	Measured Area (sqm)	% of Total	Reversibility
Fallow Land	<i>North</i>	280.81	2.46%	Reversible
Fallow Land	<i>North-West</i>	434.25	3.81%	Reversible
Fallow Land	<i>West</i>	97.26	0.85%	Reversible

Fallow Land	<i>South</i>	49.19	0.43%	Reversible
Fallow Land	<i>East</i>	924.97	8.12%	Reversible
Fallow Land	<i>Far East</i>	211.3	1.85%	Reversible
Fallow Land	<i>South-East</i>	43.68	0.38%	Reversible
Cemented	<i>AB-1 Building</i>	1006.05	8.83%	Irreversible
Cemented	<i>East Settlement</i>	376.54	3.30%	Irreversible
Green Cover	<i>Retained Vegetation</i>	7973.02	69.96%	Preserved
TOTAL	All Categories	11,397.07	100.00%	—



2.4 Pre vs Post Construction Comparison

Table 2.3 — Green Cover: Pre vs Post Construction

Parameter	Pre-Construction	Post-Construction	Net Change	Remarks
Green Cover Area	11,397.07 sqm (100%)	7,973.02 sqm (69.9%)	-3,424.05 sqm (-30.0%)	Loss to fallow + cemented
Fallow Land	0 sqm (0%)	2,041.46 sqm (17.9%)	+2,041.46 sqm	Partially reversible
Cemented / Built-up	0 sqm (0%)	1,382.59 sqm (12.1%)	+1,382.59 sqm	Irreversible
Tree Count (Area 1)	~182 trees (estimated)	102 trees (surveyed)	-80 trees (44%)	Compensatory: 800 trees
Tree Density	0.01597 trees/sqm	0.01279 trees/sqm	-0.00318 trees/sqm	Density reduced by 20%

The pre-construction baseline of 182 trees is derived by applying the current vegetative density (102 trees / 7,973.02 sqm = 0.01279 trees/sqm) to the full area of 11,397.07 sqm. This gives an original estimate of approximately 145 trees by this method; however, accounting for the denser plantation character of the eastern campus (Area 1) compared to Area 2, the adjusted estimate of 182 trees is adopted, derived by applying the overall current density (0.01597 trees/sqm) to the full area.

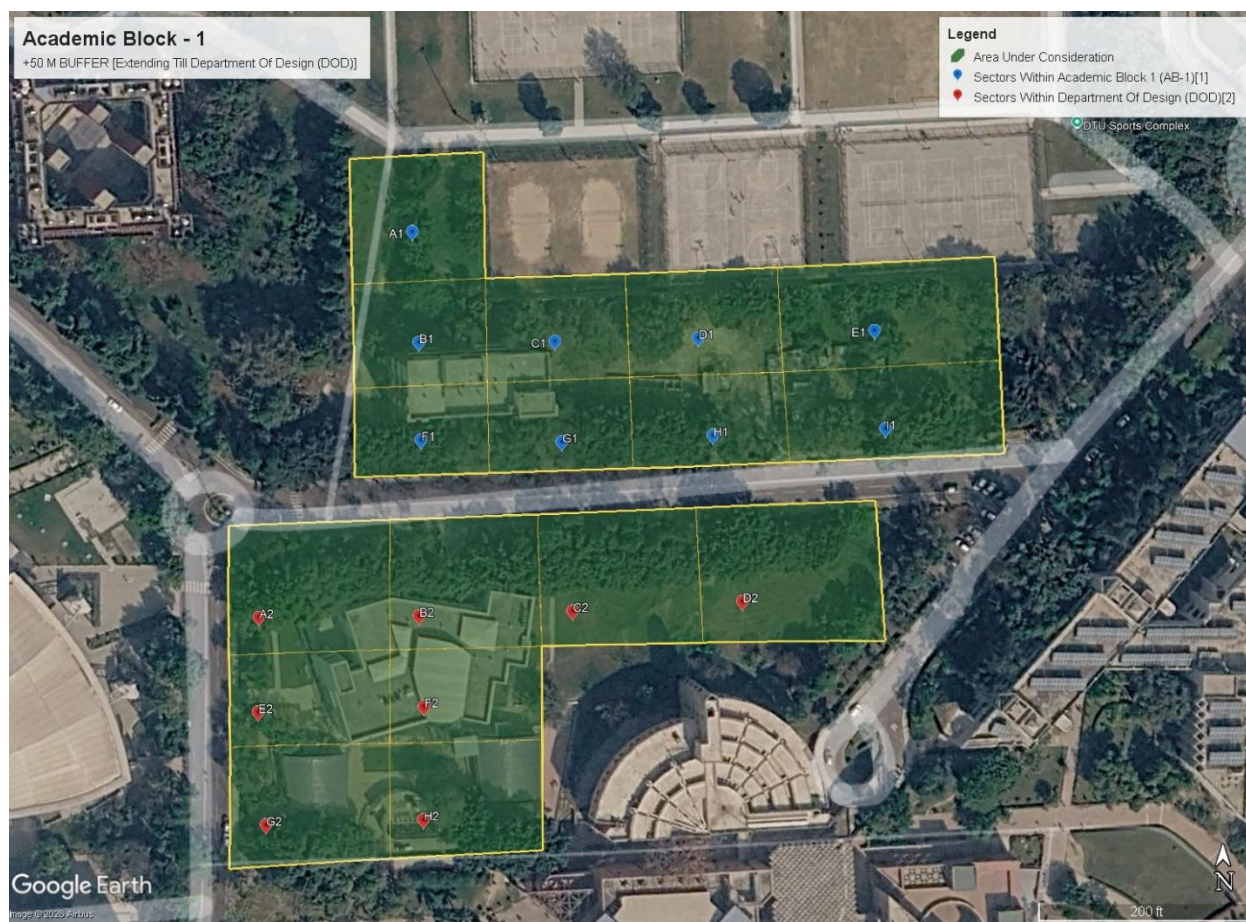
Chapter 3: Faunal Survey (Task 3)

3.1 Theoretical Background

Faunal surveys are conducted as part of ecological baseline studies to document the diversity of animal life within and around a proposed project site. In the context of EIA, fauna serves as an indicator of ecosystem health — the presence of sensitive species, pollinators, bioindicators, and endemic wildlife can significantly influence project design and mitigation requirements.

The transect method is the standard approach for urban faunal surveys. Each survey zone is treated as a transect unit, and all species encountered through direct observation, acoustic identification, or physical signs are recorded. Species are classified by taxonomic class, ecological role, and behavioural guild. Bioindicator species (dragonflies, fireflies, earthworms) signal ecosystem health, while pest and vector species indicate anthropogenic stress points.

DTU campus, with its 164-acre green expanse including a central water body, dense Banyan groves, and diverse vegetation strata, supports a rich urban wildlife community. The campus functions as a green corridor in the rapidly urbanising North-West Delhi landscape, and its ecological connectivity is directly threatened by large-scale construction such as AB-1.



3.2 Area 1 (AB-1 Grid) — Faunal Survey Results

Table 3.1 — Faunal Survey: Area 1 (Zones A1 to I1) — 25 Species

Zone	Species	Class	Role	Locality
A1	Stray Dog	<i>Mammalia</i>	Domesticated	Near campus gates & canteen
	Feral Cat	<i>Mammalia</i>	Domesticated/Opportunistic	Canteen areas, academic buildings
	Indian Palm Squirrel	<i>Mammalia</i>	Wild	Tree-lined paths, near Banyan trees
B1	Pigeon (Rock Dove)	<i>Aves</i>	Synanthropic	Rooftops, corridors of academic blocks
	Asian Koel	<i>Aves</i>	Wild	Dense canopy trees, western side
C1	Earthworm	<i>Oligochaeta</i>	Bioindicator	Moist lawns, garden beds
	Peacock	<i>Aves</i>	Wild	Open green areas — national bird
	Indian Grasshopper	<i>Insecta</i>	Wild	Grass verges, open ground
D1	Common Cricket	<i>Insecta</i>	Wild	Rocky edges, grassy margins
	Indian Honey Bee	<i>Insecta</i>	Pollinator	Hollow trees, roof eaves
E1	Fire Ant	<i>Insecta</i>	Wild	Exposed soil, tree roots
	Rhesus Macaque	<i>Mammalia</i>	Wild	Boundary wall, tall trees, water tank
	House Centipede	<i>Myriapoda</i>	Synanthropic	Basements, shaded corridors
F1	Termite	<i>Insecta</i>	Pest	Old wood fixtures, damp walls, tree stumps
	German Cockroach	<i>Insecta</i>	Pest	Canteen kitchens, drainage pipes
G1	Wood-boring Beetle	<i>Insecta</i>	Pest	Dead wood, old tree bark
	House Sparrow	<i>Aves</i>	Synanthropic	Canteen windows, garden hedges
	Rock Pigeon	<i>Aves</i>	Synanthropic	Building ledges, AC units
H1	Aedes Mosquito	<i>Insecta</i>	Vector	Stagnant water, drains
	Dragonfly	<i>Insecta</i>	Bioindicator	Campus pond, water channels
I1	Garden Lizard	<i>Reptilia</i>	Wild	Sunny walls, tree bark, rocky patches
	Common Myna	<i>Aves</i>	Synanthropic	Canteen areas, grassy patches
	House Gecko	<i>Reptilia</i>	Synanthropic	Walls, ceilings of buildings

3.3 Area 2 (DOD Grid) — Faunal Survey Results

Table 3.2 — Faunal Survey: Area 2 (Zones A2 to I2) — 24 Species

Zone	Species	Class	Role	Locality
A2	Stray Dog	<i>Mammalia</i>	Domesticated	Near campus gates & canteen
	Indian Palm Squirrel	<i>Mammalia</i>	Wild	Tree-lined paths, Banyan trees
	Oriental Magpie-Robin	<i>Aves</i>	Wild	Dense shrub edges, near water
B2	Pigeon (Rock Dove)	<i>Aves</i>	Synanthropic	Rooftops, academic block ledges
	Asian Koel	<i>Aves</i>	Wild	Dense canopy, western mango groves
	House Sparrow	<i>Aves</i>	Synanthropic	Canteen windows, garden hedges
C2	Earthworm	<i>Oligochaeta</i>	Bioindicator	Moist lawns near water body
	Red-wattled Lapwing	<i>Aves</i>	Wild	Open grassy patches near central pond
	Common Kingfisher	<i>Aves</i>	Wild	Campus water body / pond — C2-D2 boundary
D2	Indian Grasshopper	<i>Insecta</i>	Wild	Grass verges, water channel edges
	Common Cricket	<i>Insecta</i>	Wild	Rocky edges, grassy margins
	Little Cormorant	<i>Aves</i>	Wild	Rocks, branches over water body
E2	Indian Honey Bee	<i>Insecta</i>	Pollinator	Hollow trees, roof eaves
	Fire Ant	<i>Insecta</i>	Wild	Exposed soil, tree roots, western lawn
	Rhesus Macaque	<i>Mammalia</i>	Wild	Boundary wall, tall trees, water tank
F2	House Centipede	<i>Myriapoda</i>	Synanthropic	Basements, shaded corridors
	Termite	<i>Insecta</i>	Pest	Old wood, damp walls, tree stumps
	Common Tailorbird	<i>Aves</i>	Wild	Dense low bushes, ornamental hedges
H2	German Cockroach	<i>Insecta</i>	Pest	Canteen kitchens, drainage pipes
	Wood-boring Beetle	<i>Insecta</i>	Pest	Dead wood, tree stumps
	Indian Flying Fox	<i>Mammalia</i>	Wild	Tall trees near heritage buildings — dusk
I2	Aedes Mosquito	<i>Insecta</i>	Vector	Stagnant water, drains
	Common Toad	<i>Amphibia</i>	Wild	Near drains, moist ground, shaded areas
	Red-vented Bulbul	<i>Aves</i>	Wild	Fruiting trees, garden shrubs

3.4 Faunal Summary & EIA Significance

Table 3.3 — Faunal Summary by Taxonomic Class

Class	Area 1 Count	Area 2 Count	Total Species	EIA Significance
Aves (Birds)	11	10	14 (combined)	High — bioindicators; habitat fragmentation sensitive
Insecta (Insects)	9	8	11 (combined)	High — pollinators (bees), vectors (Aedes), pests (termites)
Mammalia (Mammals)	3	4	5 (combined)	High — Rhesus Macaque, Flying Fox: protected species
Amphibia	0	1	1 (Area 2)	Moderate — Toad: moist habitat indicator
Myriapoda	1	1	1 (combined)	Low — synanthropic; building-associated
Oligochaeta	1	1	1 (combined)	High — Earthworm: soil health bioindicator
TOTAL	25	24	~35 unique	Diverse urban wildlife community

The presence of Rhesus Macaque (*Macaca mulatta*) and Indian Flying Fox (*Pteropus giganteus*) — both protected under Schedule II of the Wildlife Protection Act 1972 — is of particular EIA significance. The Common Kingfisher and Red-wattled Lapwing near the campus water body indicate a healthy aquatic ecosystem. Bioindicator species (Firefly, Dragonfly, Earthworm) confirm relatively low pollution levels in the green zones, which would be adversely affected by construction dust and disturbance.

Chapter 4: Biodiversity Index Analysis (Task 4)

4.1 Theoretical Background

Biodiversity indices provide a quantitative measure of species diversity within an ecosystem, combining species richness (number of species) and evenness (distribution of individuals across species) into single metrics. In EIA practice, these indices serve as baseline benchmarks — a high pre-project diversity index strengthens the case for conservation and mitigation, while a low index may indicate an already degraded ecosystem.

The Shannon-Wiener Index (H') is the most widely used diversity metric in ecological studies and EIA reports in India. It is sensitive to both rare and common species and ranges from 0 (no diversity) to $\ln(S)$ (maximum diversity). For urban institutional ecosystems, H' values between 1.5 and 3.0 are considered moderate to good. Pielou's Evenness (E) measures how equitably individuals are distributed among species — a value close to 1.0 indicates highly even distribution. Simpson's Diversity Index ($1-D$) emphasises dominant species and is particularly useful for identifying ecosystem stability.

4.2 Species Abundance Data

The biodiversity indices are calculated from the tree census data: 15 species, 198 total individuals across the DTU campus survey area.

Table 4.1 — Species Abundance and Shannon-Wiener Calculation

S.No	Common Name	Botanical Name	Count (n_i)	$p_i = n_i/N$	$\ln(p_i)$	$p_i \times \ln(p_i)$
1	Banyan	<i>Ficus benghalensis</i>	47	0.2374	-1.4381	-0.3414
2	Peepal	<i>Ficus religiosa</i>	21	0.1061	-2.2437	-0.2380
3	Neem	<i>Azadirachta indica</i>	20	0.1010	-2.2925	-0.2316
4	Shisham	<i>Dalbergia sissoo</i>	16	0.0808	-2.5157	-0.2033
5	False Ashoka	<i>Polyalthia longifolia</i>	13	0.0657	-2.7233	-0.1788
6	Arjun	<i>Terminalia arjuna</i>	13	0.0657	-2.7233	-0.1788
7	Mulberry	<i>Morus alba</i>	11	0.0556	-2.8904	-0.1606
8	Bottle Brush	<i>Callistemon citrinus</i>	9	0.0455	-3.0910	-0.1405
9	Ber / Jujube	<i>Ziziphus mauritiana</i>	9	0.0455	-3.0910	-0.1405
10	Amaltas	<i>Cassia fistula</i>	8	0.0404	-3.2088	-0.1296
11	Gulmohar	<i>Delonix regia</i>	8	0.0404	-3.2088	-0.1296

12	Eucalyptus	<i>Eucalyptus globulus</i>	6	0.0303	-3.4965	-0.1060
13	Jamun	<i>Syzygium cumini</i>	6	0.0303	-3.4965	-0.1060
14	Sagwan/Teak	<i>Tectona grandis</i>	4	0.0202	-3.9020	-0.0788
15	Pilkhan	<i>Ficus virens</i>	7	0.0354	-3.3424	-0.1182
—	TOTAL		198	1.000	—	-2.4816

4.3 Biodiversity Indices Summary

Table 4.2 — All Diversity Indices: DTU Campus Tree Flora

Index	Symbol	Formula	Value	Interpretation
Species Richness	S	Count of species	15	Moderate — typical urban campus
Total Individuals	N	Sum of all trees	198	Good sample size
Shannon-Wiener Index	H'	$-\sum(p_i \times \ln p_i)$	2.4816	Moderate diversity — good for urban campus
Maximum Shannon (H_max)	H_max	$\ln(S) = \ln(15)$	2.7081	Theoretical maximum diversity
Pielou's Evenness	E	$H' / H_{\max}$	0.9164	Very good — species well distributed
Simpson's Index (D)	D	$\sum[n_i(n_i-1)] / N(N-1)$	0.1024	Lower = more diverse; good result
Simpson's Diversity (1-D)	1-D	$1 - D$	0.8976	High diversity; close to 1 is ideal
Reciprocal Simpson's	1/D	$1 / D$	9.77	~10 effective dominant species
Margalef's Richness	DMg	$(S-1) / \ln(N)$	2.6474	Moderate richness for site area

4.4 Interpretation & EIA Significance

Key Finding — Shannon-Wiener Index $H' = 2.4816$

DTU campus shows moderate biodiversity for an urban institutional setting. The high Evenness index ($E = 0.9164$) indicates species are well-distributed despite Banyan dominance. This qualifies the site as ecologically sensitive under MoEFCC guidelines, warranting biodiversity-conscious site planning.

The Shannon-Wiener Index of 2.48 falls within the moderate diversity range (1.5–3.5), which is commendable for an urban institutional campus in North-West Delhi. The Pielou's Evenness of 0.9164 (out of a maximum of 1.0) is particularly noteworthy — it indicates that trees are distributed fairly evenly across species, meaning no single species completely dominates to the exclusion of others (despite Banyan being the most abundant at 23.7%).

Simpson's Diversity ($1-D = 0.8976$) further confirms the high ecological value of the campus green cover. A value approaching 1.0 indicates that if two trees are randomly selected, there is nearly a 90% probability they belong to different species — a strong indicator of biodiversity.

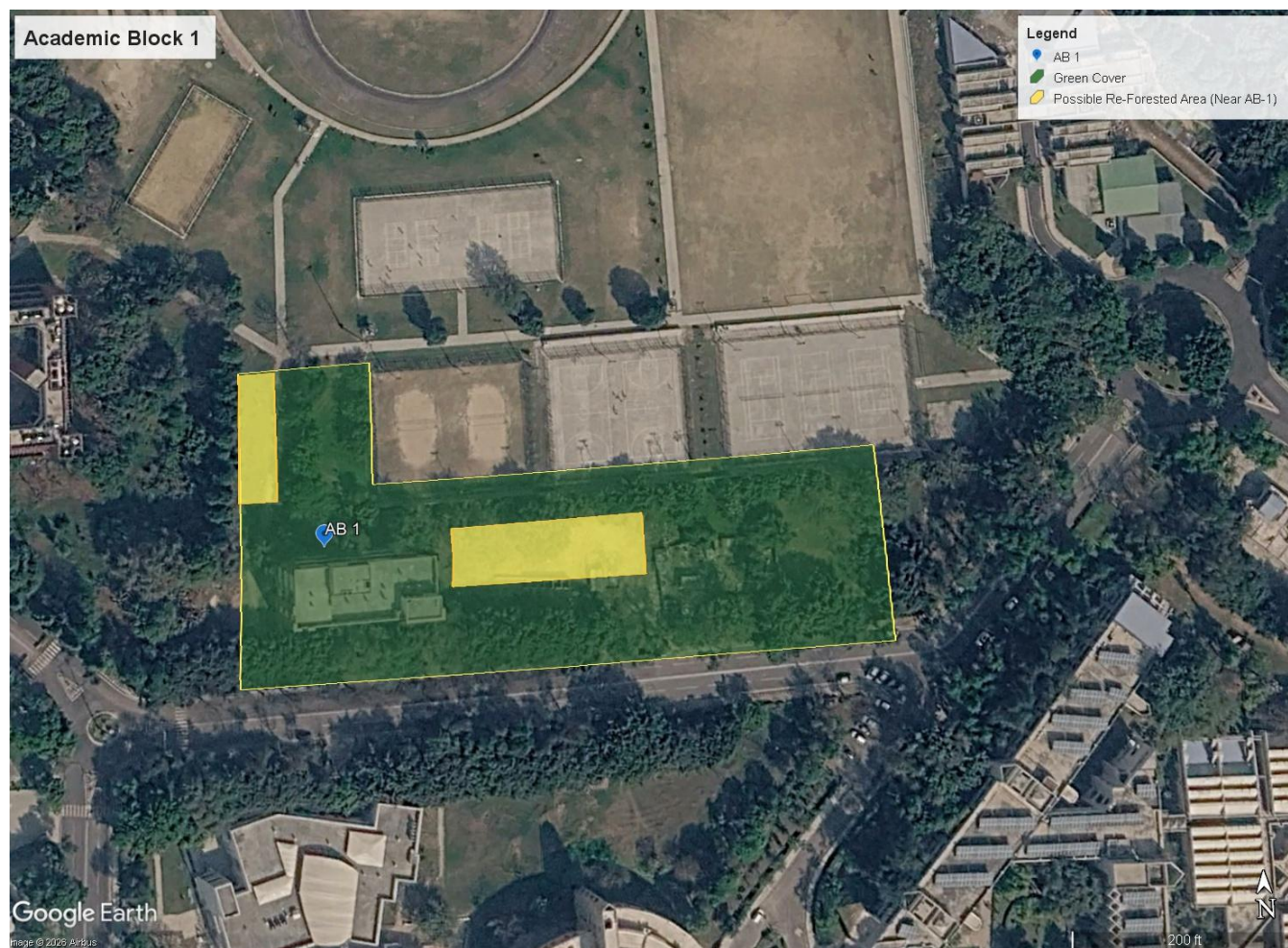
For EIA purposes, this baseline H' value of 2.48 must be maintained or improved post-construction. The compensatory afforestation plan (Chapter 5) has been designed specifically to introduce species that will raise the H' index through increased species richness and improved evenness.

Chapter 5: Green Belt Design & Mitigation Plan (Task 5)

5.1 Theoretical Background

Ecological mitigation in EIA follows a hierarchical approach: first Avoid, then Minimize, then Restore, and finally Offset (AMRO hierarchy). The green belt is both a mitigation and restoration measure — a continuous band of vegetation planted around a construction or built-up area that buffers ecological impacts, filters dust, reduces noise, provides wildlife habitat, and improves the visual character of the built environment.

As per the DTU campus policy and MoEFCC Compensatory Afforestation Guidelines, every tree lost must be compensated by planting 10 trees (1:10 ratio). The compensatory plantation must prioritise native species — trees that are naturally adapted to the Delhi climate and support local wildlife. The handbook (EN302 Survey Camp) specifically mandates Neem, Peepal, and Arjun as priority species for the green belt around AB-1.



5.2 Applicable Laws — Green Belt & Compensatory Afforestation

Table 5.1 — Legal Provisions for Mitigation & Green Belt

Law / Policy	Provision	Implication for AB-1 Mitigation
Delhi Preservation of Trees Act, 1994	Compensatory plantation mandatory; minimum 1:10 ratio for trees felled in urban areas	800 trees must be planted on DTU campus within 6 months of tree removal
MoEFCC Compensatory Afforestation Guidelines	Native species priority; plantation to be monitored for 5 years; survival rate >75%	Neem, Peepal, Arjun as priority species; DTU Estate Office responsible for monitoring
NGT Order on Urban Green Cover (2019)	Green belt mandatory around construction sites; minimum 3m width	3m native green belt around full AB-1 perimeter; no bare ground within buffer zone
EN302 Handbook — Group 6 Task 5	Minimum 3m green belt; native species: Neem, Peepal, Arjuna; landscape architecture proposal	Green belt design included in this chapter with spacing, species, and monitoring plan
Wildlife Protection Act, 1972 (Schedule II)	Rhesus Macaque and Indian Flying Fox protected; habitat disturbance must be minimised	Construction activities near Macaque habitats (western boundary) must avoid peak activity hours
National Building Code 2016	Minimum 33% open/green space on site	Post-construction landscaping within AB-1 precinct; rooftop greening recommended

5.3 Mitigation Hierarchy

Table 5.2 — AMRO Mitigation Hierarchy

Level	Strategy	Specific Measure	Target / Indicator
AVOID	Minimize footprint	Optimise AB-1 layout; retain Banyan trees (GBH >56 in) within 5m; no-go buffer zone around heritage trees	Zero heritage trees lost; GPS tagging of all large trees before construction
AVOID	Pathway realignment	GIS-based review; reroute paths to avoid root zones of retained trees	20% reduction in pathway-related tree loss
MINIMIZE	Transplantation	Transplant trees with GBH <25 in from footprint to green belt zones before clearing	Min. 10 trees transplanted; survival rate >70% at 3 months
MINIMIZE	Root protection zone	Root barriers + compaction guards within 2m radius of all 102 retained trees	Zero root damage; monthly inspection log maintained
MINIMIZE	Phased construction	Avoid simultaneous disturbance of full footprint; retain soil and vegetation where possible	50% reduction in simultaneous clearing area
RESTORE	Fallow revegetation	Revegetate all 2,041.46 sqm of fallow zones with native grasses, shrubs, ground cover	80% green cover restored within 12 months post-construction
RESTORE	Green belt — AB-1	3m wide native green belt around full AB-1 perimeter: Neem, Peepal, Arjun (5×5m spacing)	Full perimeter green belt established within 18 months
OFFSET	Compensatory planting	800 trees planted campus-wide (1:10 ratio); ~200 on-site in retained green, ~600 in other DTU zones	800 trees; survival >75% Year 1; >85% Year 3
OFFSET	Biodiversity enhancement	Introduce Jamun, Amaltas, Gulmohar to raise H' index above baseline (2.48)	H' index recalculated annually; target ≥ 2.48

5.4 Compensatory Afforestation Plan

Based on the loss of 80 trees at a 1:10 replacement ratio, 800 trees are required. The plan prioritises native species (60%) as mandated by the EN302 handbook. The total space required for the plantation is approximately 2.83 hectares (with 30% buffer for access paths), which represents only 4.26% of DTU's total 164-acre campus — making it entirely feasible within existing campus green zones.

Table 5.3 — Compensatory Afforestation Plan: 800 Trees, 8 Species

S.No	Species	Botanical Name	Category	No. to Plant	Spacing (m)	GBH @5yr (in)	% Total
1	Neem	<i>Azadirachta indica</i>	Native Priority	200	5×5	18–22	25%
2	Peepal	<i>Ficus religiosa</i>	Native Priority	160	6×6	20–28	20%
3	Arjun	<i>Terminalia arjuna</i>	Native Priority	120	5×5	22–30	15%
4	Jamun	<i>Syzygium cumini</i>	Native Support	80	5×5	15–20	10%
5	Amaltas	<i>Cassia fistula</i>	Native Support	64	4×4	12–16	8%
6	Gulmohar	<i>Delonix regia</i>	Native Support	64	5×5	10–14	8%
7	Shisham	<i>Dalbergia sissoo</i>	Native Support	64	4×4	18–24	8%
8	Pilkhan	<i>Ficus virens</i>	Supplementary	48	7×7	25–35	6%
—	TOTAL		8 Spp	800	—	—	100%

5.5 Space Requirement Analysis

Table 5.4 — Afforestation Space Requirement (800 Trees)

Species	Trees	Spacing (m)	Area/Tree (sqm)	Raw (sqm)	Area + 30% Buffer (sqm)	% of Required	Total
Neem	200	5×5	25	5000	6500	23.0%	
Peepal	160	6×6	36	5760	7488	26.5%	
Arjun	120	5×5	25	3000	3900	13.8%	
Jamun	80	5×5	25	2000	2600	9.2%	
Amaltas	64	4×4	16	1024	1331	4.7%	
Gulmohar	64	5×5	25	1600	2080	7.4%	
Shisham	64	4×4	16	1024	1331	4.7%	
Pilkhan	48	7×7	49	2352	3058	10.8%	
TOTAL	800	—	—	21,760 sqm	28,288 sqm	100%	

Space Availability Analysis

Total space required: 28,288 sqm (2.83 ha) | Available in Area 1 retained green (7,973 sqm): ~220 trees (27%)
 | Off-site on DTU campus required: ~580 trees (73%) | DTU total campus: 663,708 sqm | Space needed = 4.26% of campus — highly feasible across avenue planting, open grounds, and boundary belts.

5.6 Monitoring & Compliance Schedule

Table 5.5 — Monitoring Plan: Pre, During & Post Construction

Phase	Activity	Parameter Target	Frequency	Responsible / Standard
Pre-Construction	Tree tagging & mapping	100% of 102 trees + 50m buffer GPS tagged	Once (Month 0)	DTU EO + Group 6 MoEFCC EIA Guidelines
Pre-Construction	Transplantation	Trees GBH <25 in; survival >70% at 3 months	Once (Month 1)	Certified horticulturist Delhi Tree Act 1994
Construction	Tree protection audit	Fencing + root guard integrity; zero violations	Weekly	Site Engineer IS 11384:1985
Construction	Dust monitoring	PM10 <100 µg/m ³ at site boundary	Fortnightly	Contractor + Group 3 NAAQS 2009; C&D Rules 2016
Post-Construction	Compensatory planting	800 trees planted as per species plan	Phased (Month 19–24)	DTU Horticulture Delhi Preservation of Trees Act 1994
Post-Construction	Survival check rate	Survival >75% Year 1; >85% Year 3; replace dead saplings	Quarterly	DTU Horticulture MoEFCC Comp. Afforestation Guidelines
Post-Construction	Biodiversity index	H' Shannon index recalculated; target H' ≥ 2.48	Annual (Year 1,3,5)	DTU Env. Dept. CBD + MoEFCC Guidelines
Post-Construction	Green assessment belt	Min. 3m width; >80% canopy cover around AB-1	Bi-annual	DTU Estate Office NGT Order 2019; NBC 2016
Post-Construction	Carbon ledger	Net positive carbon balance by Year 5	Annual	DTU Env. Dept. IPCC Guidelines + MoEFCC

6. Conclusions & Recommendations

6.1 Summary of Findings

The ecological baseline study of the AB-1 site and surrounding campus area at DTU has yielded a comprehensive picture of the green cover and biodiversity status prior to and post proposed construction. Key findings are summarised below:

- 198 trees of 15 species enumerated across 17 survey zones; Banyan (*Ficus benghalensis*) is the dominant species with 47 individuals (23.7%)
- Shannon-Wiener Diversity Index $H' = 2.4816$ — moderate diversity; Pielou's Evenness $E = 0.9164$ — very good distribution
- Green cover loss: 3,424.05 sqm (30%) of Area 1 converted to fallow/cemented land; estimated 80 trees lost from pre-construction baseline of ~182
- 49 faunal species recorded across both areas including 2 Schedule II protected species (Rhesus Macaque, Indian Flying Fox) and multiple bioindicator species confirming ecosystem health
- Compensatory afforestation requirement: 800 trees (1:10 ratio) across 8 native species, requiring ~2.83 ha; feasible within DTU's 164-acre campus

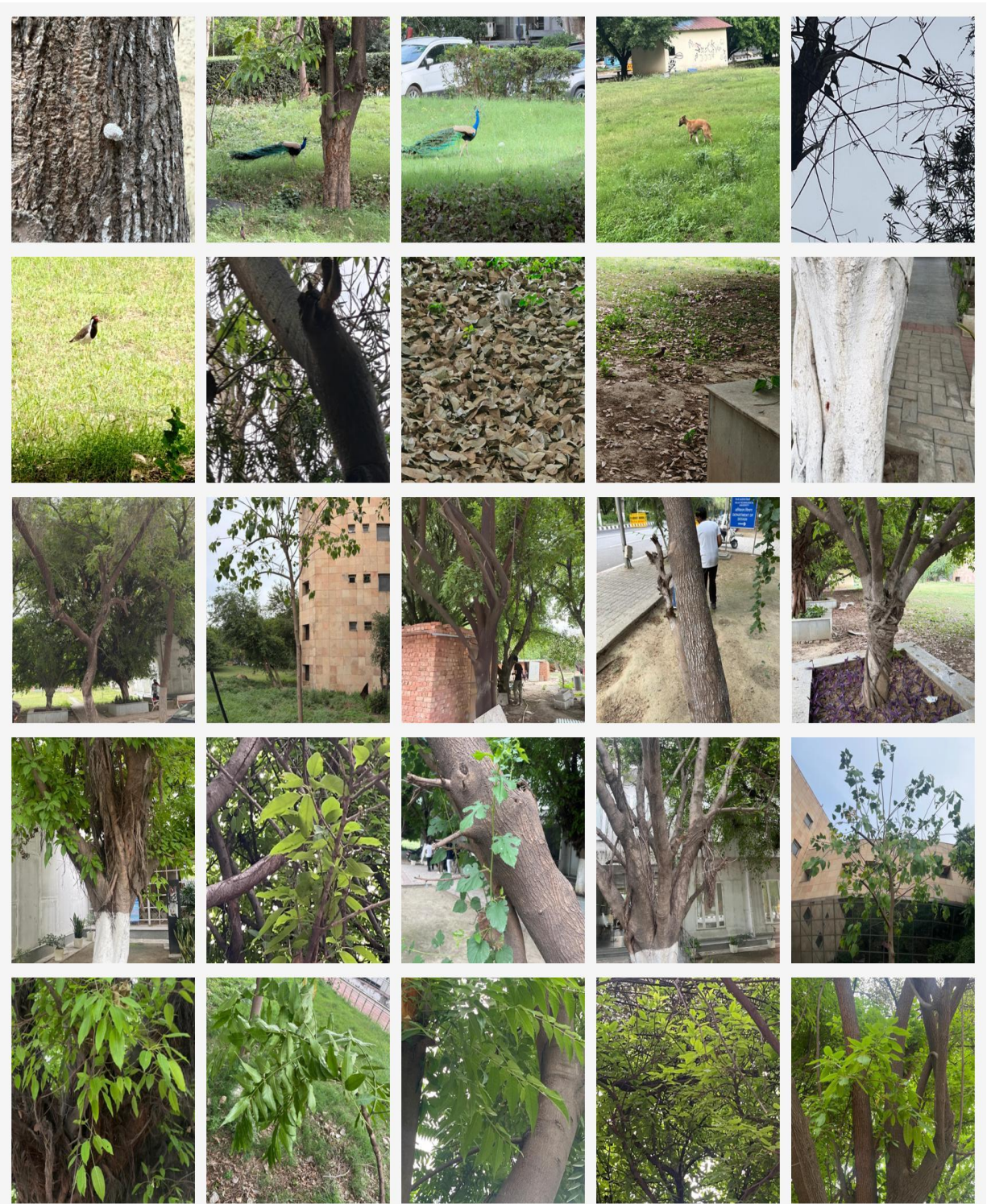
6.2 Recommendations

- Immediate GPS tagging and protective fencing of all 102 retained trees before construction activities commence
- Transplant all trees with GBH <25 inches from the construction footprint to the proposed green belt zones
- Strictly enforce the 3m no-disturb zone around heritage Banyan trees (GBH >56 inches)
- Initiate compensatory plantation of 800 trees within 6 months of any tree removal, prioritising Neem, Peepal, and Arjun as per DTU policy and handbook mandate
- Establish a 3m wide native green belt around the full AB-1 perimeter within 18 months of construction completion
- Conduct annual biodiversity index recalculations to monitor H' recovery against the baseline of 2.48
- Protect the campus water body (C2-D2 zone) from construction runoff — critical habitat for Kingfisher, Lapwing, Cormorant, Dragonfly
- Avoid construction activities near the western boundary (E1/E2 zones) during peak Macaque activity hours (06:00–09:00 and 17:00–19:00 hrs)

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